

How DNS Works: Translating Human Intent to Machine Reality

When you type a website name into your browser, you are using a human-friendly address. Computers, however, cannot read names; they communicate strictly using numerical IP addresses. The Domain Name System (DNS) is the underlying infrastructure that translates the domain name you typed into the exact IP address of the server hosting that website.

Here is the exact sequence of events that occurs in the milliseconds after you hit enter:

The Recursive Resolver: Your browser first asks a server called a recursive resolver (usually provided by your internet service provider) for the website's IP address. If the resolver does not already have the answer saved in its recent memory (cache), it begins the search.

The Root Nameserver: The resolver contacts a root nameserver. The root server does not know the exact IP address, but it reads the end of your domain (such as .app or .com) and directs the resolver to the specific server that handles those top-level extensions.

The TLD Nameserver: The resolver then asks the Top-Level Domain (TLD) nameserver. For a .app domain, this server knows exactly who manages your specific website's records and points the resolver to the final destination.

The Authoritative Nameserver: The resolver reaches the authoritative nameserver. This is the master ledger that holds the definitive, final records for your specific domain.

The Response: The authoritative server locates the exact IP address (known as an A Record) and hands it back to the recursive resolver. The resolver delivers it to your browser, allowing your browser to connect directly to the host server and load the website.

What is a CNAME Record?

While an A Record points a domain directly to an IP address (numbers), a CNAME (Canonical Name) Record acts as an alias that points a domain to another domain name.

For example, when hosting on a platform like Vercel, the IP addresses of their servers might occasionally change to balance server loads. If you bought ayyanhassan.com, pointing it directly to a static Vercel IP address could break your site if Vercel updates their infrastructure. Instead, you use a CNAME record to point www.ayyanhassan.com to ayyanhassan-portfolio.vercel.app. This tells the DNS system: "Whenever someone looks for my custom domain, seamlessly redirect them to wherever the Vercel app is currently hosted"